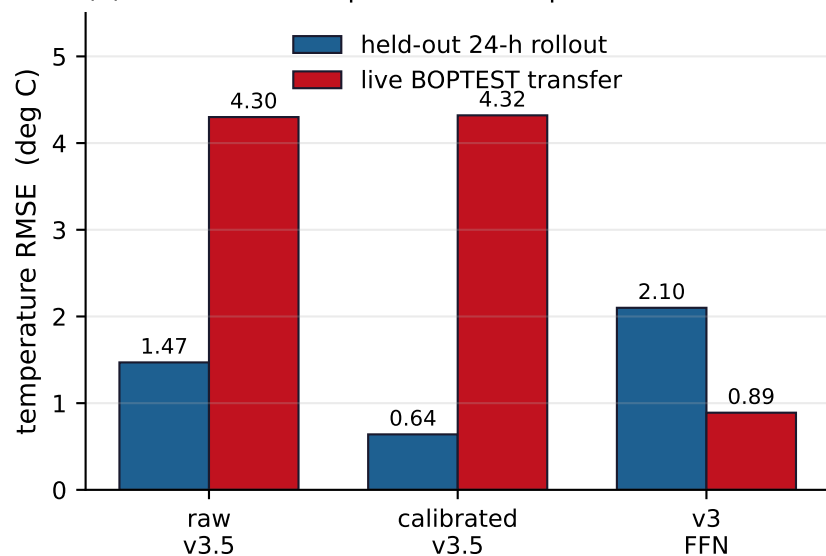


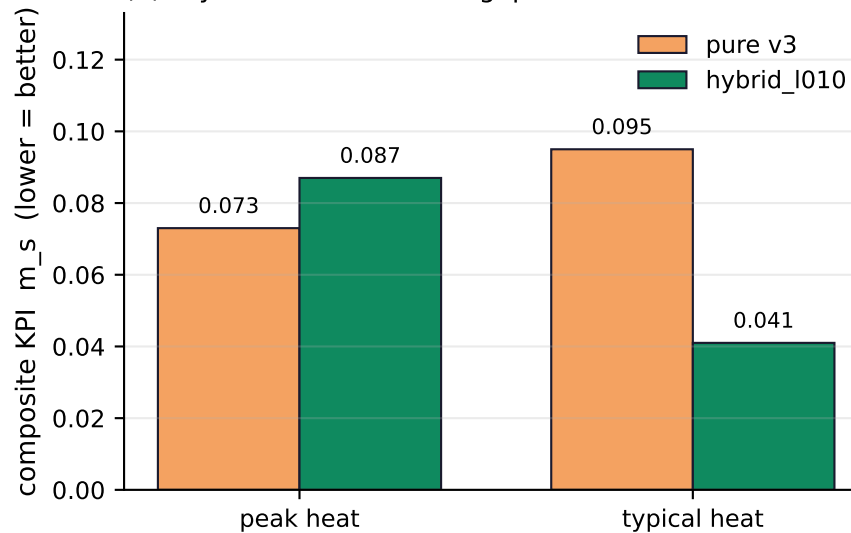
Calibrated physical twin as a soft regulariser for HVAC reinforcement learning

BOPTEST bestest_air (source) + 3 hydronic testcases (transfer) — hybrid backend 85x faster than RTE HTTP-Docker

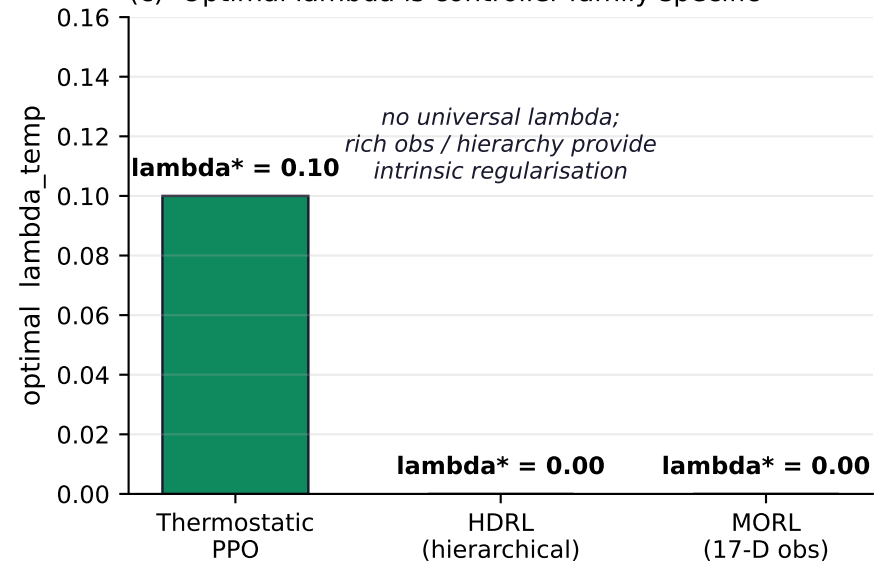
(a) The calibration paradox: best predictive twin = worst RL env



(b) Hybrid loss closes the gap on thermostatic PPO

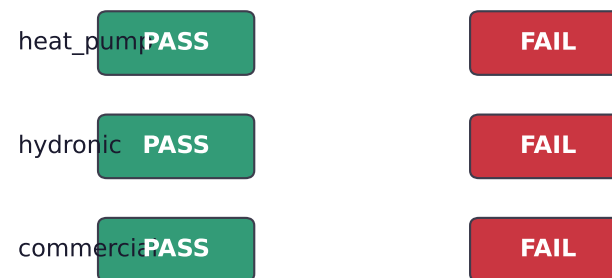


(c) Optimal lambda is controller-family specific



(d) Component-level decomposition across BOPTEST hydronic family (Block 3, N = 3)

Surrogate component (C_zon ratio: 1.91 +/- 0.08) Controller component (Frozen direct-TSup policy)



Surrogate transfers structurally / Controller transfers only when action interface matches